

Intelligent Association



Presented by

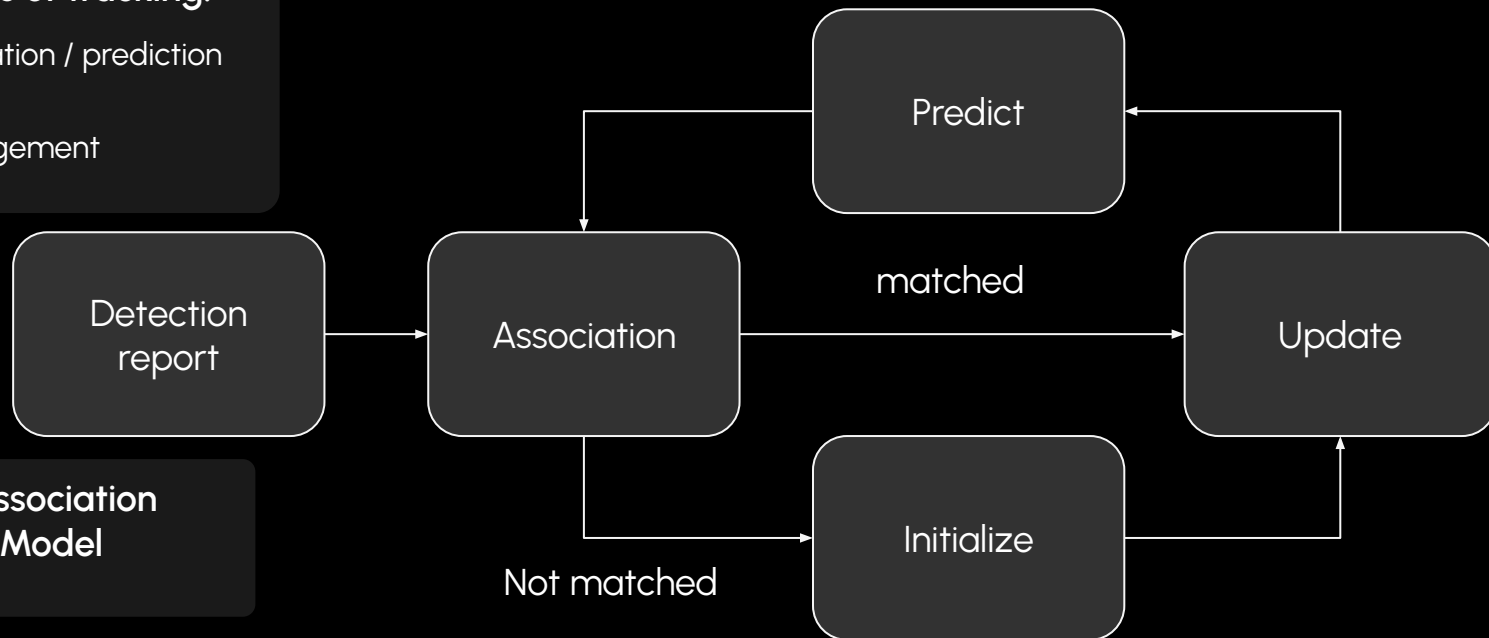
Spencer

Randolph

Background - Tracking

Three main parts of tracking:

- State estimation / prediction
- Association
- Track Management



**GOAL: Improve Association
Through Learned Model**

Milestone 1:

Data Generation

Milestone 2:

Tracking Architect
Model Training

Milestone 3:

Model Training
Evaluation

Data generation goals

Training and validation require large amounts of data

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graph TD; A[Training and validation require large amounts of data] --> B[Realistic: Important for data to represent as close to real world as possible]; A --> C[Repeatable: data should be seeded and repeatable for testing]; A --> D[Labeled: The data needs to be labeled for training as well as contain ground truth]; A --> E[Controllable: data presets with varying degrees of challenge];
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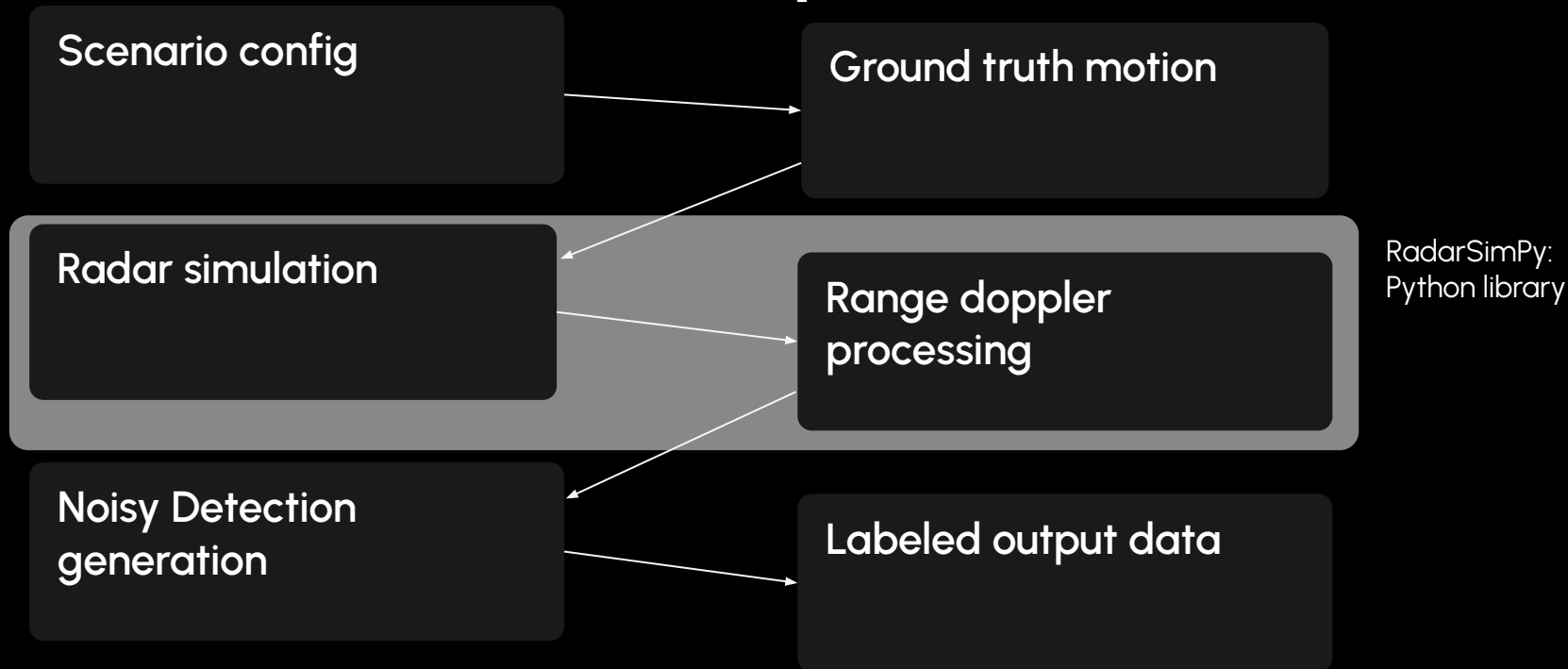
Realistic: Important for data to represent as close to real world as possible

Repeatable: data should be seeded and repeatable for testing

Labeled: The data needs to be labeled for training as well as contain ground truth

Controllable: data presets with varying degrees of challenge

Data Generation Pipeline:



RadarSimPy:

zhengyu peng

- PHD EE Texas Tech

Est 2021

Python based radar
simulation software



Other:

- MATLAB radar toolbox
- CARLA
- Open RCS

Used by:

- Universities
- Government & national labs
- Defence industry
- Automotive industry

FMCW Radar:

Carrier freq: 77 ghz

BandWidth: 750 mhz

Range resolution: 0.2 m

- $C / (2 * \text{BandWidth})$

Chirp time: 60 microseconds

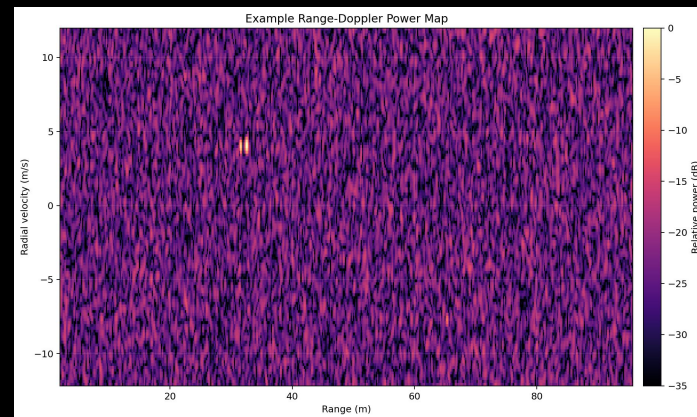
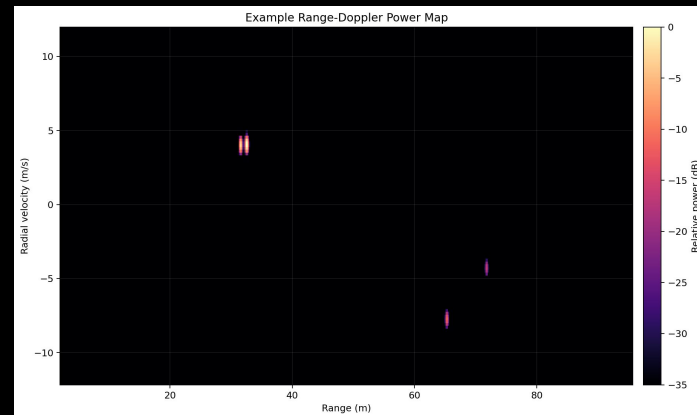
Pulses per frame: 64

Velocity resolution: 0.4 m/s

- $\text{wavelength} / (2 * \text{coherent processing time})$

Range doppler map: 128 x 512

Max range: 100m



Configs

Easy:

Targets: 4

Detection prob: 96%

False alarms: 0.8

Near-target FA: 0.15

Occlusion chance: 0.1%

Maneuver chance: 0.6%

Medium:

Targets: 6

Detection prob: 88%

False alarms: 2.5

Near-target FA prob: 0.45

Occlusion chance: .4%

Maneuver chance: 1.5%

Hard:

Targets: 9

Detection prob: 76%

False alarms: 5.5

Near-target FA prob: 0.9

Occlusion chance: 0.9%

Maneuver chance: 3%

Occlusion: when one object might block part of the view of the second object

Chance for long term drop (missed detections)

synthetic
solution

Maneuver: when an object change in motion breaks signify off its predicted path

Chance of random acceleration spike positive or negative

Scenarios

- Used to mimic patterns seen in real world use cases
- Only controls the start
- Each data set uses a random scenario

Random:
Random start with random direction
Focuses:

- General tracking

Crossing:
Starts at edge of fov, directed towards a center point
Focuses:

- Target Identity swaps

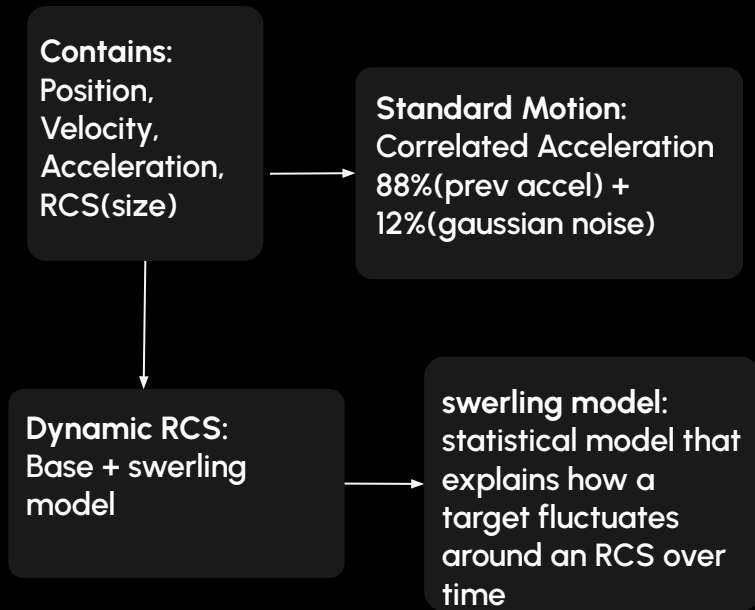
Dense:
Starts targets clumped
Random direction
Focuses:

- Target Identity

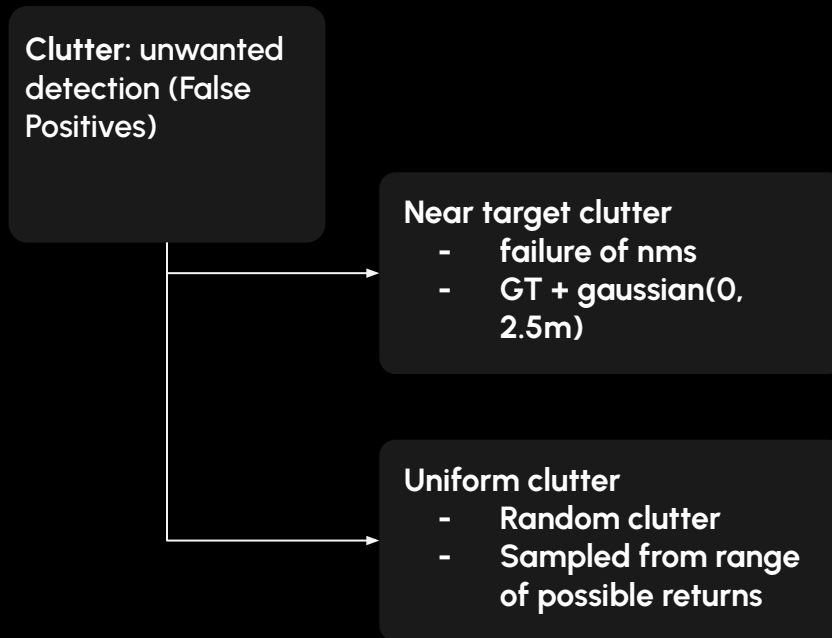
Parallel:
Starts targets near, similar motion
Focuses:

- Target Identity

Targets



Clutter



Feature Extraction: hybrid approach

Noise: introduced into ground truth range, azimuth and radial velocity. Then flows down

range:
True range + sampled noise

Azimuth:
True azimuth + sampled noise

Radial velocity:
True radial velocity + sampled noise

Gaussian: mean 0
Std $f(x)$

$f = 0.30 + 0.004 * \text{range}$

$f = 0.28 \text{ deg} + 0.010 \text{ deg} * \text{range}$

$f = 0.28 \text{ m/s}$

- Calculated: geometric features calculated from ground truth plus noise

Position:
 $\hat{x} = rX + nR * \cos(nA)$
 $\hat{y} = rY + nY * \sin(nA)$

RadarSimPy Simulated:
radar based features pulled from range doppler map using noisy bin

Power: strength of targets radar return

Detection Report: Labeled output data

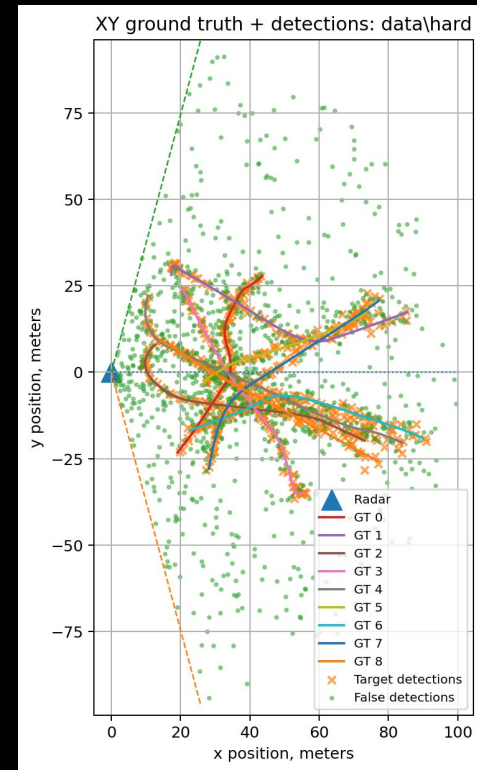
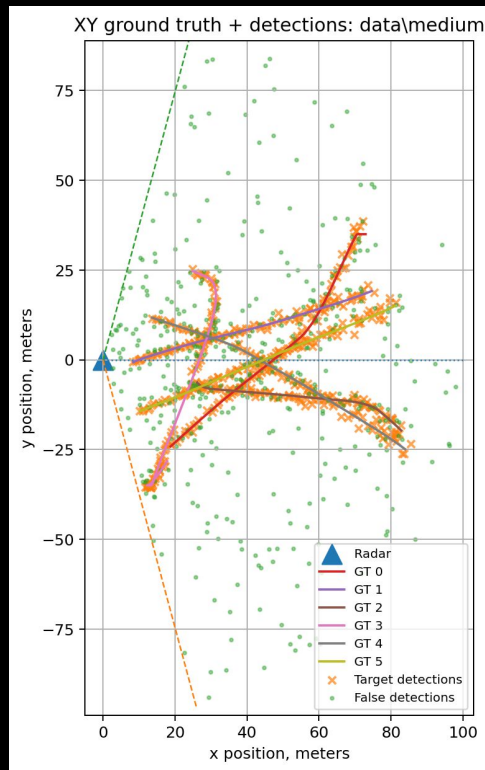
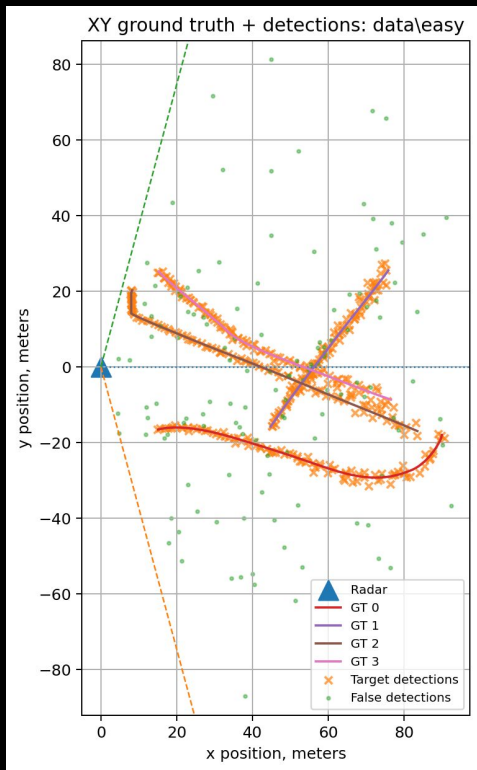
Ground truth:

- Frame id
- Target id
- X
- Y
- Vx
- Vy
- Range
- Azimuth
- Radial velocity
- Rcs mean
- Swerling model
- In fov

Detections:

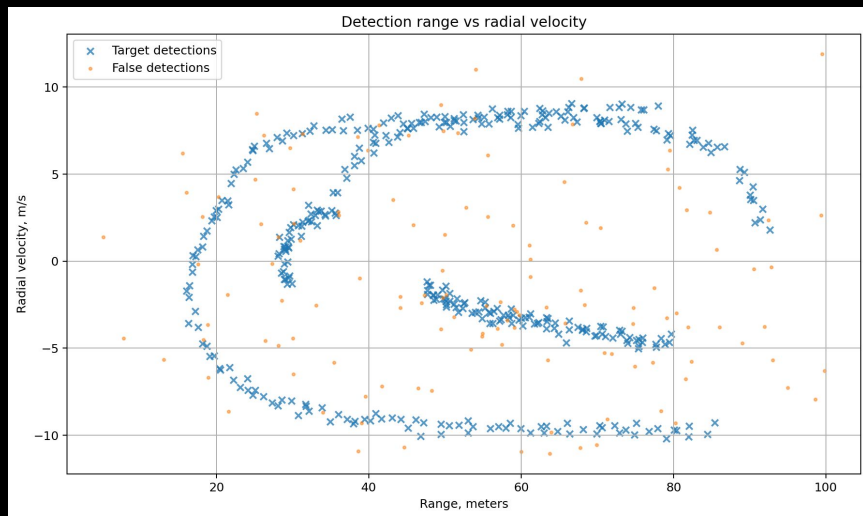
- Frame id
- Detect id
- Target id
- X
- Y
- Radial velocity
- Power

Results: overview

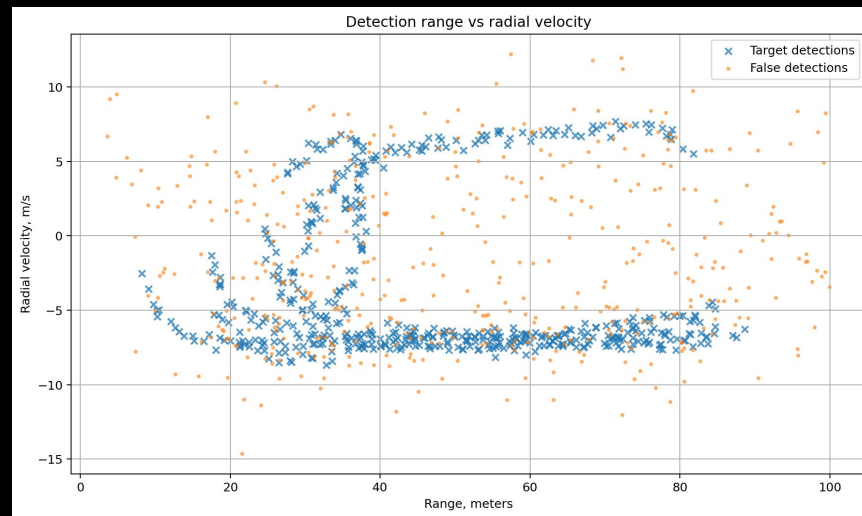


Results: Range vs Radial Velocity

Easy

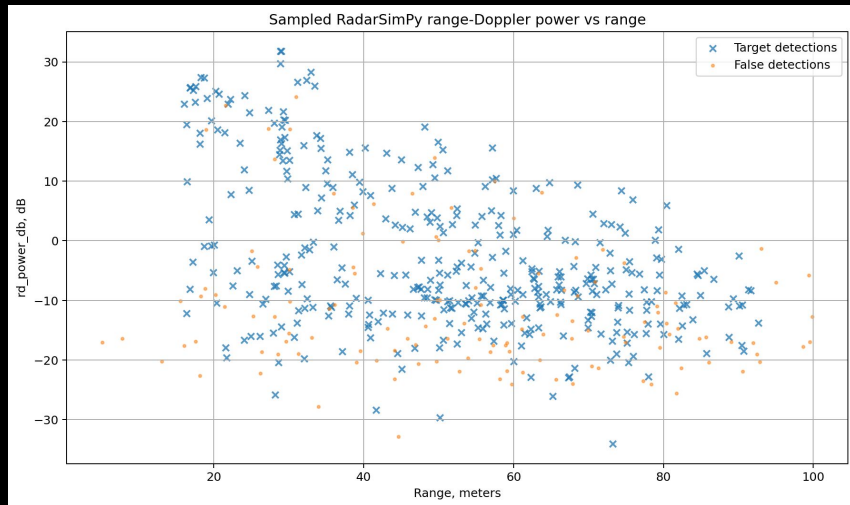


Medium

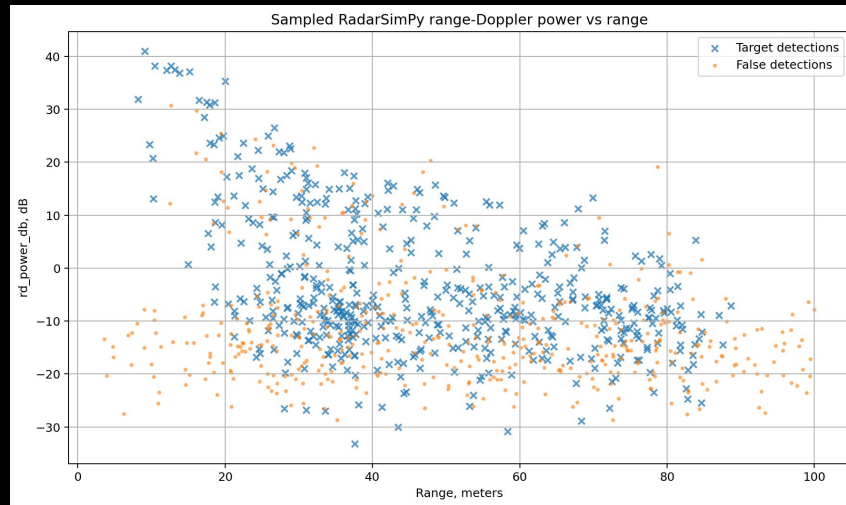


Results: power vs range

Easy

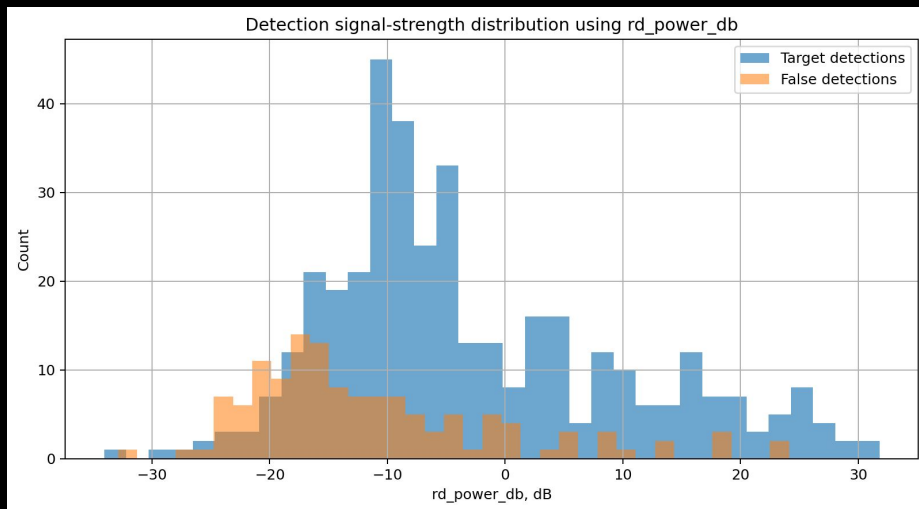


Medium

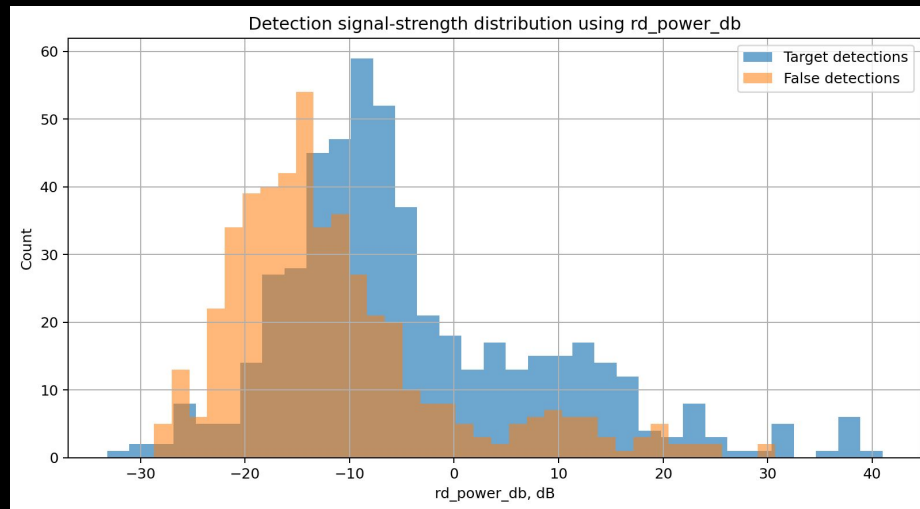


Results: power distribution

Easy

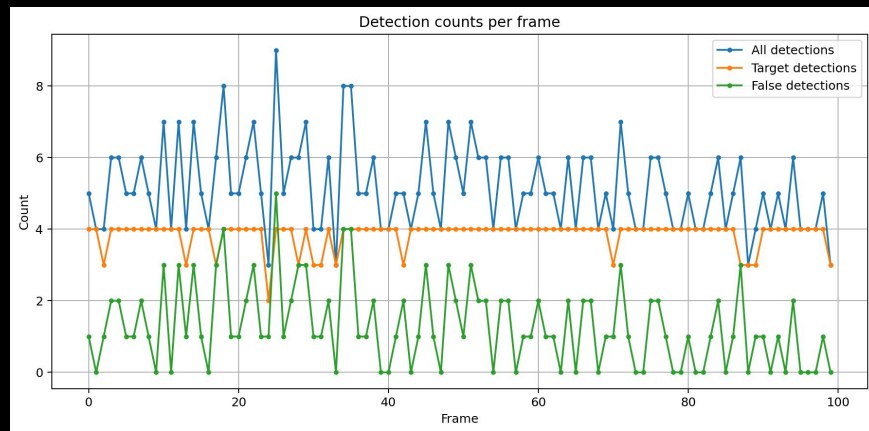


Medium

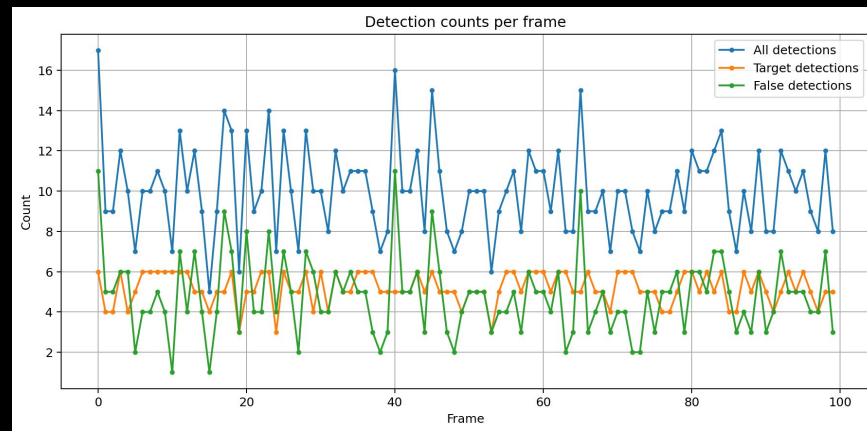


Results: detections per frame

Easy

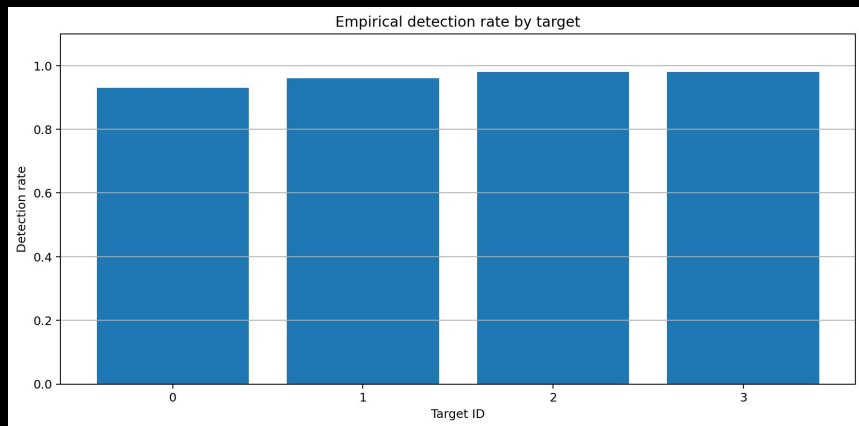


Medium

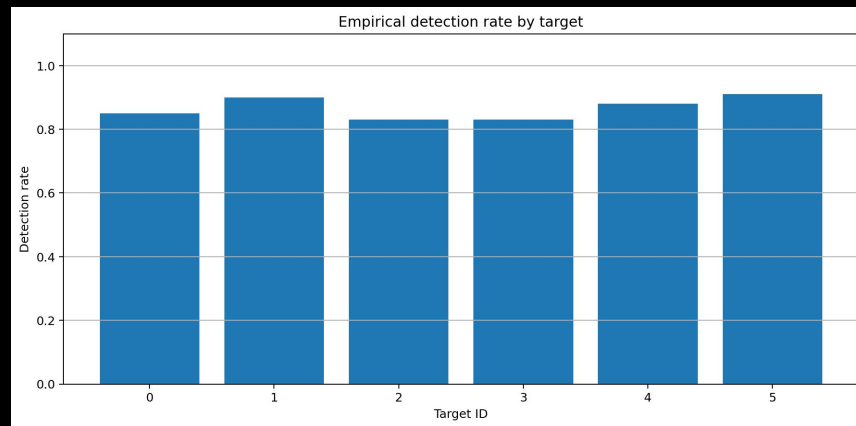


Results: detections rate

Easy

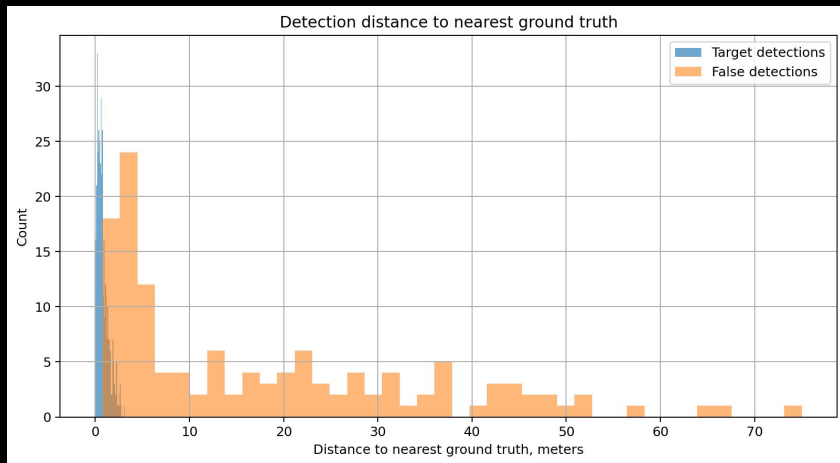


Medium

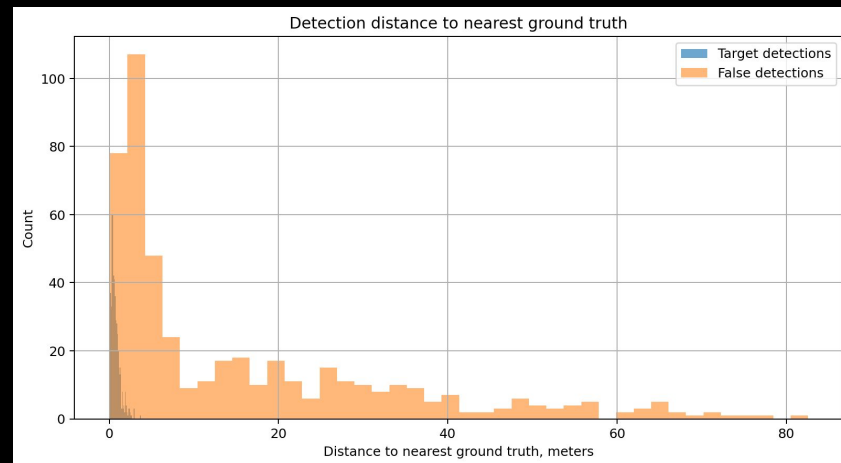


Results: clutter distance

Easy

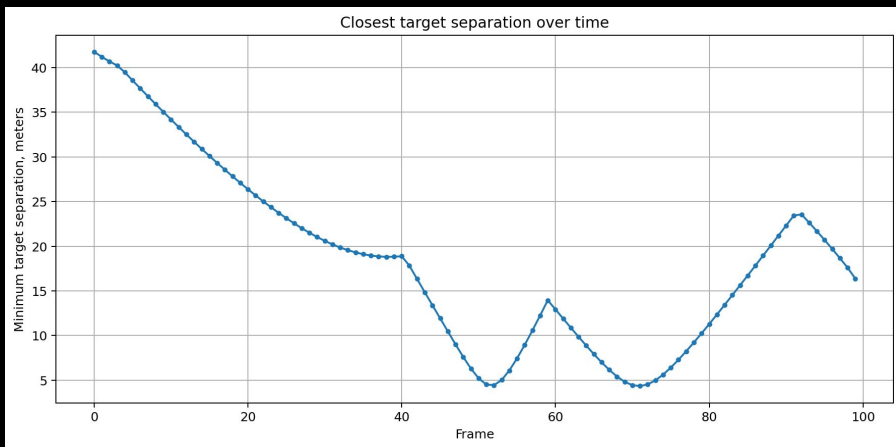


Medium

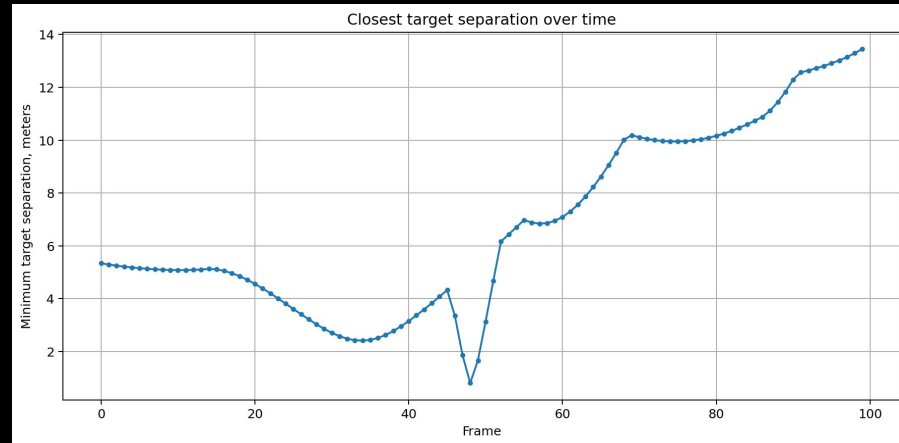


Results: closest clutter

Easy



Medium



Limitations:

Point based simulation:

- No object size, shape or orientation
- No true geometric occlusion

Clutter:

- False alarms are injected statistically
- Uniform clutter may be too weak

Potential solutions (**costly**):

Mesh based simulation:

- object size, shape or orientation
- geometric occlusion

Full Environment:

- Introduce more realistic clutter

None Hybrid approach:

- Full processing including CFAR

Questions?